

Basic Cloud Security models-39

Introduction

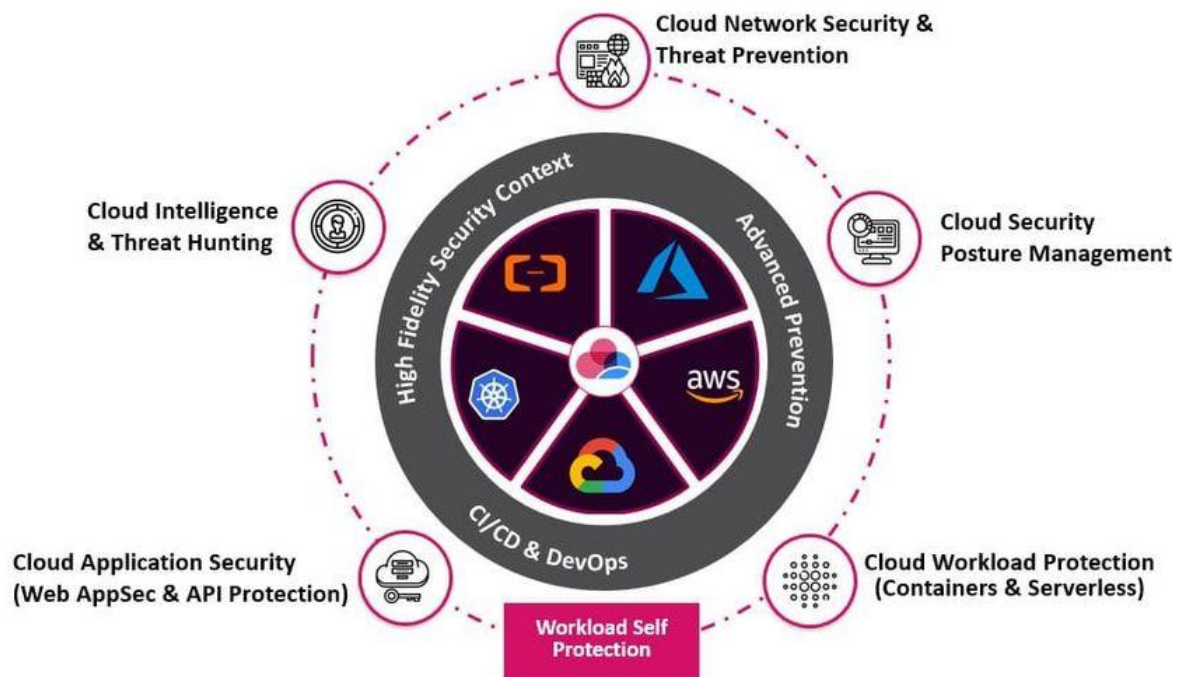


Cloud security models define how responsibilities are shared between the provider and the customer. They ensure confidentiality, integrity, and availability of data and services. These models are critical because cloud environments are multi-tenant and accessible over the internet.

Core Cloud Security Models

- Shared Responsibility Model
 - Security duties are divided between the provider and the customer.
 - Provider secures the infrastructure, physical hardware, and core services.
 - Customer secures operating systems, applications, and user access.
 - Example: In AWS, the provider manages data centers, networking, and hardware; the customer manages data, encryption, and IAM.
- Identity and Access Management (IAM)
 - Controls who accesses what resources.
 - Includes authentication (proving identity) and authorization (granting access).
 - Uses role-based access control (RBAC) and multi-factor authentication (MFA).

- Reduces insider threats and prevents unauthorized access.
- Encryption Model
 - Data encrypted at rest (storage) and in transit (network).
 - Providers offer key management services (KMS).
 - Customers manage encryption keys in some cases.
 - Example: TLS for secure data transfer, AES for storage encryption.
- Zero Trust Security Model
 - “Never trust, always verify” approach.
 - Every access request is authenticated and authorized regardless of origin.
 - Relies on continuous monitoring and micro-segmentation.
 - Protects against lateral movement inside networks.
- Defense in Depth Model
 - Security is layered across multiple levels: physical, network, application, and data.
 - Each layer compensates if another fails.
 - Example: firewalls, IDS/IPS, access controls, endpoint protection.
- Security by Design Model
 - Security integrated during system development, not added later.
 - Includes secure coding practices, regular code audits, and DevSecOps.
 - Reduces vulnerabilities in cloud-native applications.
- Compliance and Regulatory Model
 - Ensures that services follow industry standards.
 - Certifications like ISO 27001, SOC 2, HIPAA, and GDPR compliance.
 - Providers supply compliance-ready infrastructure; customers ensure workloads follow policies.



Key Examples from Providers

- AWS Shared Responsibility Model documentation clarifies boundaries.
- Microsoft Azure enforces Zero Trust using Conditional Access policies.
- Google Cloud encrypts all data at rest by default with AES-256.

Challenges in Applying Cloud Security Models

- Misunderstanding boundaries of responsibility.
- Weak IAM policies leading to privilege escalation.
- Poor key management practices.
- Delays in patching workloads by customers.
- Lack of monitoring and auditing.

Best Practices

- Apply least privilege access with IAM roles.
- Enable MFA for all accounts.
- Encrypt sensitive data both at rest and in transit.
- Use provider security services like AWS GuardDuty, Azure Security Center.

- Regularly review compliance reports and perform security audits.
- Train employees on cloud security responsibilities.

Cloud security models provide a structured way to manage risks in a shared environment. By applying IAM, encryption, zero trust, and compliance-focused practices, organizations strengthen their security posture. The success of these models depends on continuous monitoring, clear responsibility boundaries, and active participation from both cloud providers and customers.